Allen Moncey Varghese

Smart Manufacturing · Machine Learning · Digital Twins · Process Engineering
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Manufacturing Engineering Ph.D. candidate (ASU, 4.0 GPA, expected 2027) who builds AI systems for real factories: deep learning on high-frequency machine sensor data, computer-vision quality inspection, digital twins in Unreal Engine and NVIDIA Omniverse, and hands-on metal additive manufacturing with a robot-mounted Meltio DED head. Combines industrial engineering training (DOE, SPC, Lean Six Sigma Green Belt) with production experience that cut cycle time 25% across 15 projects.

EDUCATION

Ph.D., Manufacturing Engineering — Aug 2023 – Present
Arizona State University, Tempe, AZ · GPA 4.0/4.0

M.S., Industrial Engineering (Industrial Statistics) — Aug 2020 – Aug 2022
Arizona State University, Tempe, AZ · GPA 3.78/4.0 · Thesis: Multi-Variant Spatially Informed Rapid Testing for Epidemic Model

B.Tech., Mechanical Engineering — Jul 2015 – May 2019
Vellore Institute of Technology, India

RESEARCH EXPERIENCE

Graduate Research Associate, AI & Digital Twin Manufacturing — Aug 2023 – Present
Arizona State University (Dr. Binil Starly & Dr. Ashif Iquebal), Tempe, AZ

- Developed the FC-VQ-VAE architecture, compressing high-frequency sensor streams (22 kHz CNC audio, 10 kHz accelerometer) into discrete codebook tokens at ~25× compression for real-time digital-twin synchronization under edge-hardware and bandwidth limits; published in Manufacturing Letters (2026).
- Designed and trained a complex-valued convolutional autoencoder for CNC machinery audio generation from complex STFT inputs (reconstruction MSE ≈ 0.20) for virtual-factory and VR operator-training environments; published in Manufacturing Letters (2025).
- Built an end-to-end machining-condition classifier on the compressed tokens, showing they retain enough information for process monitoring and anomaly detection.
- Engineered an AI-driven defect-detection system for bandsaw tooth-quality inspection with industry partner M.K. Morse Co., replacing manual inspection with an automated computer-vision pipeline and a quantitative wear metric.
- Run metal additive manufacturing research on a wire-laser directed energy deposition (DED) cell: a Meltio deposition head on a FANUC R-1000iA/80F industrial robot with a 2-axis positioner; diagnosed and fixed integration faults between the Meltio software and the robot controller.
- Designed and curated a 4-hour CNC audio dataset using DOE across feed rate and depth of cut, a reproducible benchmark for machining digital-twin models.

Graduate Research Volunteer, Digital Twin Simulation — Nov 2022 – Aug 2023
Arizona State University (Dr. Ashif Iquebal), Tempe, AZ

- Created 3D CAD models of machining centers in Fusion 360 and integrated them into Unreal Engine 5.1 for real-time digital-twin experiments; simulated tool-failure scenarios in VR to inform failure prevention.

Graduate Research Assistant, Epidemic Modeling & Data Analytics — Dec 2020 – Jul 2022
Arizona State University (Dr. Giulia Pedrielli), Tempe, AZ

- Built a multi-variant SEIR epidemic simulator with testing and vaccination dynamics in MATLAB (dde23), achieving NMSE ≈ 0.28 with AR(2) calibration for Omicron and Delta; built Tableau pipelines for data cleaning and model validation.

INDUSTRY EXPERIENCE

Production and Operations Engineer — May 2019 – Jul 2020

Hebron Greenfield, Kochi, India

- Reduced production cycle time by 25% and increased profit margins by 20% across 15 projects through data-driven scheduling and process improvement.
- Achieved 95% on-time delivery through systematic purchase-order planning and expediting.
- Implemented a raw-material reuse policy, reducing waste by 10%.

May 2017 – Jul 2017

Manufacturing Intern

Kerala Agro Machinery Corporation Ltd, Kochi, India

- Reduced diesel tiller assembly time by 15% through shift-overlap optimization; produced technical reports on new prototypes.

PUBLICATIONS

- Varghese, A., Starly, B., & Iquebal, A. (2026). Vector quantization of complex sensor data in limited network bandwidth and edge hardware constrained digital twins. *Manufacturing Letters*, 49, 1550–1561. doi:10.1016/j.mfglet.2026.08.060
- Varghese, A., Starly, B., & Iquebal, A. (2025). Complex value autoencoder for machinery audio generation. *Manufacturing Letters*, 44, 1468–1478. doi:10.1016/j.mfglet.2025.06.166
- Azad, F. T., Dodge, R. W., Varghese, A. M., et al. (2022). SIRTEM: Spatially informed rapid testing for epidemic modeling and response to COVID-19. *ACM Transactions on Spatial Algorithms and Systems*, 8(4), 1–43.

TECHNICAL SKILLS

AI & Deep Learning: PyTorch, VQ-VAE, Convolutional Autoencoders, GANs, Diffusion Models, Transformers, scikit-learn, Anomaly Detection

Signal Processing: Complex STFT, Wavelet Decomposition, EMD, FFT Analysis, Audio DSP

Computer Vision: Contour Extraction, Defect Detection, Quality Inspection, Image Segmentation

Digital Twin & Simulation: Unreal Engine 5, NVIDIA Omniverse, Real-Time Sensor Streaming, Edge Deployment

Additive Manufacturing & Robotics: Metal Directed Energy Deposition (Meltio wire-laser), Robotic Deposition Cells, Robot–Software Integration

Manufacturing & Quality: DOE, Six Sigma (DMAIC), SPC, Root Cause Analysis, SolidWorks, Fusion 360, ANSYS, AutoCAD

Programming & Data: Python, MATLAB, SQL, R, C++, Pandas, NumPy, Tableau, JMP Pro

HONORS & CERTIFICATIONS

- Fulton Fellows Fellowship: full Ph.D. tuition + 4-year research stipend, ASU (2023–2027) · University Graduate Fellowship, ASU (2026–2027)
- NSF Travel Award, NAMRC 53 (2025) · Engineering Graduate Fellowship, Ira A. Fulton Schools of Engineering (2020–21)
- Lean Six Sigma Green Belt, KPMG India